

Suggested problems: 1-13

Problem 1: Suppose that X is a continuous random variable with probability density function

$$f(x) = \begin{cases} cx^2 & : 0 \leq x \leq 2, \\ 0 & : \text{otherwise,} \end{cases}$$

where c is a constant.

(a) Compute c .

(b) Compute $P(X \geq 1)$.

(c) Compute $P(X \geq 1 \mid X \geq 0.5)$.

Problem 2: Suppose that the probability density function of a random variable X is given by

$$f(x) = \begin{cases} \frac{1}{8}x & : 0 \leq x \leq 4, \\ 0 & : \text{otherwise.} \end{cases}$$

After a random value of X is observed, let Y be the integer closest to X . Find the probability function of Y .